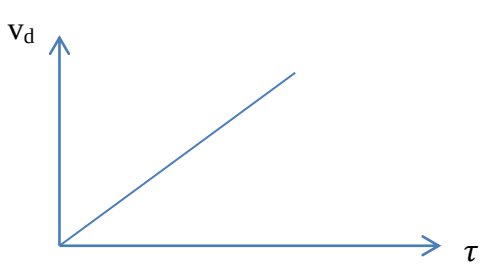
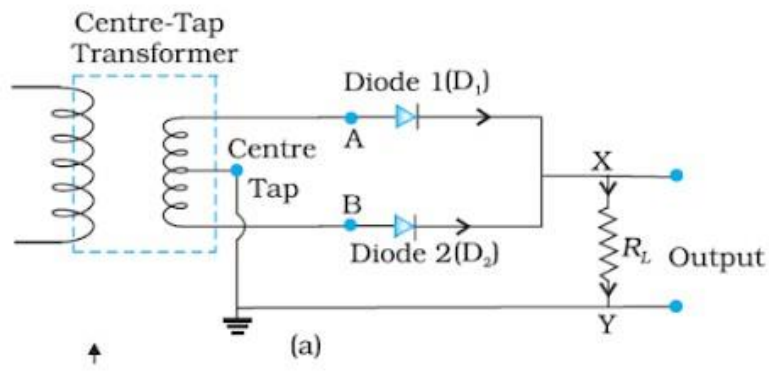
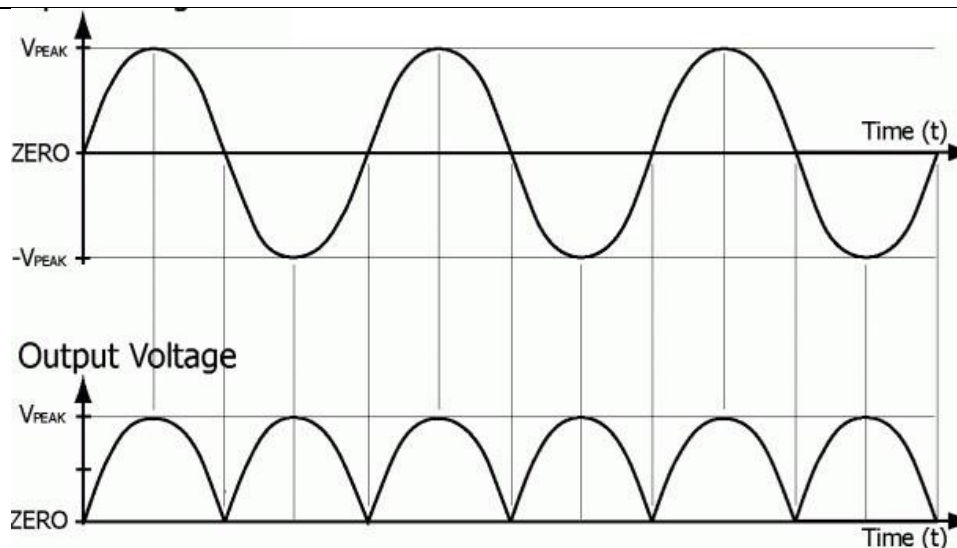


S.NO	SECTION-A	MAR KS
1	a	1
2	b	1
3	c	1
4	d	1
5	d	1
6	c	1
7	a	1
8	c	1
9	a	1
10	b	1
11	d	1
12	c	1
13	(D) Battery is disconnected from the capacitor. So, $Q = \text{Const.}$ Energy = $Q^2 / 2C = Q^2 d / 2 \epsilon_0 A$. so energy $\propto d$	1
14	(D)	1
15	(A)	1
16	(C)	1
	<u>SECTION B</u>	
17	(I) <i>Drift velocity $\propto$ Relaxation time</i>  (ii) Alternating current changes direction every half cycle. So average drift velocity is zero	1 $\frac{1}{2}$ $\frac{1}{2}$

18	Here $r = 5 \text{ cm} = 0.05 \text{ m}$, $I_1 = 10 \text{ A}$, $I_2 = 5 \text{ A}$, $l = 20 \text{ cm} = 0.2 \text{ m}$, $F = ?$ Since $F = \frac{\mu_0}{4\pi} \frac{2I_1 I_2}{r} l$ $\therefore F = \frac{10^{-7} \times 2 \times 10 \times 5 \times 0.2}{0.05}$ or $F = 4 \times 10^{-5} \text{ N}$ The direction of F acting on conductor Y is in the plane of the paper, perpendicular to Y and is directed away from conductor X.	$\frac{1}{2}$ $\frac{1}{2}$ 1
19	$a = 0.1 \text{ mm} = 10^{-4}$ $\lambda = 6000 \times 10^{-10} = 6 \times 10^{-7}$ $D = 0.5 \text{ m}$ for 3rd dark $\sin \theta = 3\lambda$ $\sin \theta = 3\lambda / a = x / D$ $x = 3\lambda D / a$ $= 3 \times 6 \times 10^{-7} \times 0.5 / 10^{-4}$ $= 9 \text{ mm}$	1 $\frac{1}{2}$ $\frac{1}{2}$
20	(a) S, W, X (b) Reason for low binding energy: In heavier nuclei, the Colombian repulsive effects can increase considerably and can match/ offset the attractive effects of the nuclear forces. This can result in such nuclei being unstable.	1 1
21	 (a)	1



$\frac{1}{2} + \frac{1}{2}$

OR

1. Forward Bias: (i) Within the junction diode the direction of applied voltage is opposite to that of built-in potential. (ii) The current is due to diffusion of majority charge carriers through the junction and is of the order of milli amperes. (iii) The diode offers very small resistance in the forward bias.
2. Reverse Bias: (i) The direction of applied voltage and barrier potential is same. (ii) The current is due to leakage of minority charge carriers through the junction and is very small of the order of μA . (iii) The diode offers very large resistance in reverse bias.

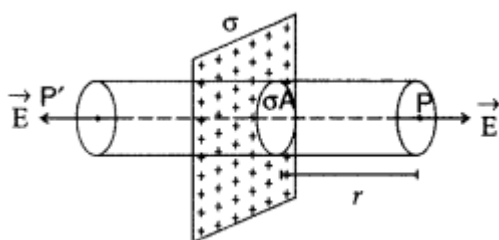
1

1

[SECTION – C]

22

Let σ be the surface charge density (charge per unit area) of the given sheet and let P be a point at distance r from the sheet where we have to find $E \rightarrow$



Choosing point P', symmetrical with P on the other side of the sheet, let us draw a Gaussian cylindrical surface cutting through the sheet as shown in the diagram. As at the cylindrical part of the Gaussian surface, $E \rightarrow$ and $dS \rightarrow$ are at a right angle, the only surfaces having $E \rightarrow$ and $dS \rightarrow$ parallel are the plane ends

$\frac{1}{2}$

$$\begin{aligned}\therefore \phi_E &= \oint \vec{E} \cdot d\vec{S} + \oint \vec{E} \cdot d\vec{S} \\ &= \oint E dS + \oint E dS = EA + EA = 2EA\end{aligned}$$

1/2

...[As E is outgoing from both plane ends, the flux is positive.

This is the total flux through the Gaussian surface.

Using Gauss' law, $\phi_E = \frac{q}{\epsilon_0}$

$$\therefore 2EA = \frac{q}{\epsilon_0} = \frac{\sigma A}{\epsilon_0} \quad \dots[\text{As } q = \sigma A]$$

$$\therefore E = \frac{\sigma}{2\epsilon_0}.$$

1/2

This value is independent of r.

(b) The electric field $E = \sigma / 2 \epsilon_0$ is constant

1/2

Work done $w = fd$

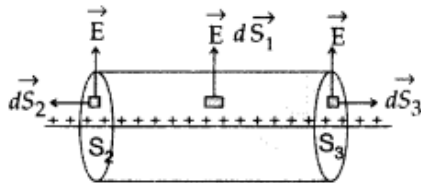
$$\therefore w = qEd = \frac{\sigma qd}{2\epsilon_0}$$

OR

Gauss's law in electrostatics : It states that "the total electric flux over the surface S in vacuum is $1/\epsilon_0$ times the total charge (q)."

1

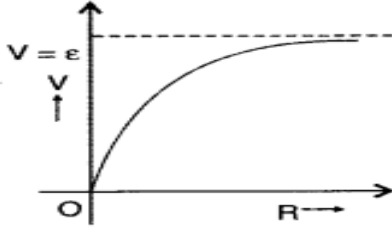
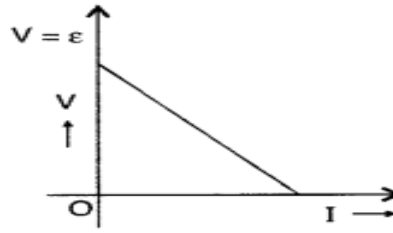
Contained in side S $\therefore \phi_E = \oint_S \vec{E} \cdot d\vec{S} = \frac{q}{\epsilon_0}$

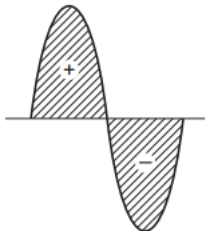
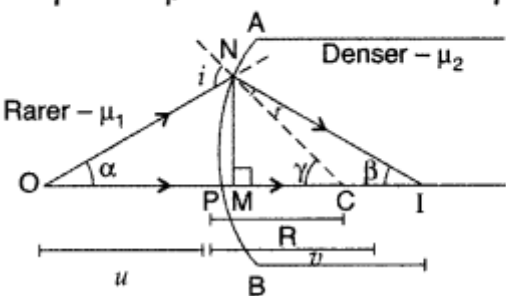


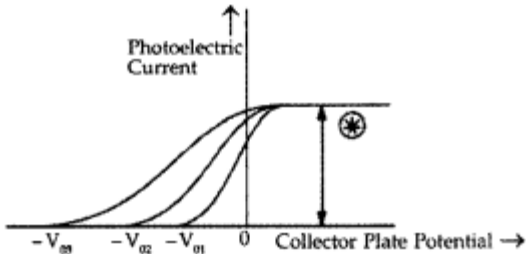
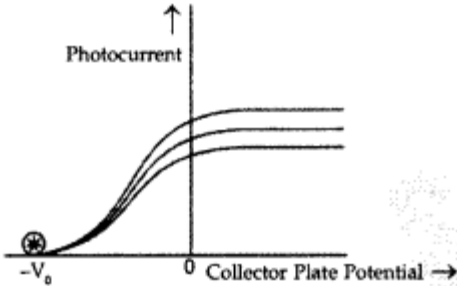
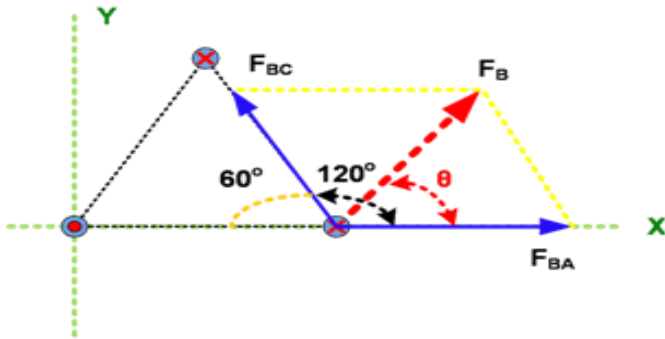
1/2

Electric field due to an infinitely long straight wire : Consider an infinitely long straight line charge having linear charge density λ to determine its electric field at distance r . Consider a cylindrical Gaussian surface of radius r and length l coaxial with the charge. By symmetry, the electric field E has same magnitude at each point of the curved surface S_1 and is directed radially outward.

1/2

	Total flux through the cylindrical surface, $\oint \vec{E} \cdot d\vec{S} = \oint_{s_1} \vec{E} \cdot d\vec{S}_1 + \oint_{s_2} \vec{E} \cdot d\vec{S}_2 + \oint_{s_3} \vec{E} \cdot d\vec{S}_3$ $= \oint_{s_1} E dS_1 \cos 0^\circ + \oint_{s_2} E dS_2 \cos 90^\circ + \oint_{s_3} E dS_3 \cos 90^\circ$ $= E \oint dS_1 = E \times 2\pi r l$ As λ is the charge per unit length and l is the length of the wire, so charge enclosed is, $q = \lambda l$ By Gauss's theorem : $\oint \vec{E} \cdot d\vec{S}_1 = \frac{q}{\epsilon_0} \Rightarrow E \times 2\pi r l = \frac{\lambda l}{\epsilon_0}$ $\therefore E = \frac{\lambda}{2\pi\epsilon_0 r}$	$\frac{1}{2}$ $\frac{1}{2}$
23	(a) Graph between V and I  (b) Graph between V and R  (c) $I = \frac{E}{R+r}$ $I = \frac{E}{4+r} \Rightarrow E = 4 + r \dots(i)$ Also $0.5 = \frac{E}{9+r} \Rightarrow E = 4.5 + 0.5r \dots(ii)$ From equations (i) and (ii) $4 + r = 4.5 + 0.5r \therefore r = 1\Omega$ Using this value of r, in equation (i) we get $E = 5V$	1+1 1
24	(a) (i) $\tau = MB \sin \theta$ $\theta = 90^\circ$ $\tau = 1.5 \times 0.22 \times \sin 90^\circ$ $\tau = 1.5 \times 0.22 \times 1 = 0.33 \text{ Nm}$ (ii) $\theta = 180^\circ$ $\tau = 1.5 \times 0.22 \times \sin 180^\circ = 0$ (b)(i) 0° (ii) 180°	1 1 $\frac{1}{2}$ $\frac{1}{2}$
25	Ans.(a) We know that Power = $P = VI$ that is curve of power will be having maximum	1

	amplitude, equals to multiplication of amplitudes of voltage (V) and current (I) curve. So, the curve will be represented by A. (b) As shown by shaded area in the diagram, the full cycle of the graph consists of one positive and one negative symmetrical area.  Hence, average power over a cycle is zero. (c) As the average power is zero, hence the device may be inductor (L) or capacitor (C) or the series combination of L and C	1 1
26	- e.m. waves in the wavelength range 10 nm to 10^{-3} nm are X-rays. - X-rays are generated by bombarding a metal target with high energy electrons. - Uses : 1. Diagnosis of bone fractures. 2. Treatment of some forms of cancer.	1 1 $\frac{1}{2}$ $\frac{1}{2}$
27	In $\triangle NOC$, i is an exterior angle, $\therefore i = \alpha + \gamma$ Similarly, from $\triangle NIC$, we have $\gamma = r + \beta \Rightarrow r = \gamma - \beta$  Suppose all the rays are paraxial Then the angles i, r, α, β and γ will be small	

	 (d) Graph for different intensities : 	1
	SECTION D	
29	((i) Ans. (d) Reason: in reverse biased p-n junction l, potential difference across a junction becomes $[V+V_b]$ (ii) Ans. (c) concentration of positive and negative charges near the junction (iii) Ans. (c) Reason: D2 is reverse biased and D1 is forward biased. So, 2phm and 4 ohm are in series and connected to 12 V. Hence, $I = 12 / (3+4) = 1.71A$ (iv) Ans. (d) Si and Ge are semiconductors Or Ans. (b)	1 mark for each correct option
30	1.(c) 2. (b) 3. (b) 4.(d) OR (a)	
	SECTION E	
31	(i) <u>In</u> the figure of forces acting on conductor B with the angles  Finding F_{BA} :	

$$F_{BA} = \frac{\mu_o i_b i_a l}{2\pi a_{ba}}$$

$$F_{BA} = \frac{(4\pi \times 10^{-7})(1)(1)(1)}{2\pi(1)} = 2 \times 10^{-7} \text{ Newton}$$

Finding F_{BC} :

$$F_{BC} = \frac{\mu_o i_b i_c l}{2\pi a_{bc}}$$

$$F_{BC} = \frac{(4\pi \times 10^{-7})(1)(1)(1)}{2\pi(1)} = 2 \times 10^{-7} \text{ Newton}$$

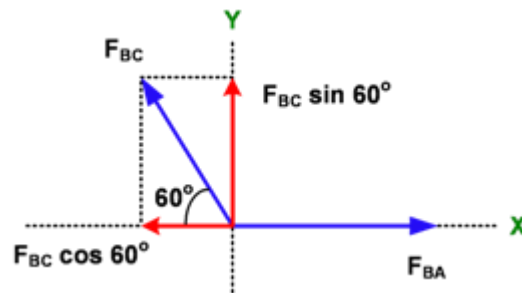
resultant named F_B :

$$F_B = \sqrt{(F_{BA})^2 + (F_{BC})^2 + 2(F_{BA})(F_{BC}) \cos 120^\circ}$$

Insert your data :

$$F_B = 2 \times 10^{-7} \text{ Newton}$$

b) Using vectors components to get the direction of the resultant :



$$\begin{aligned}\Sigma F_X &= F_{BA} - F_{BC} \cos 60^\circ \\ \Sigma F_X &= (2 \times 10^{-7}) - (2 \times 10^{-7})\left(\frac{1}{2}\right) = 1 \times 10^{-7} \text{ N} \\ \Sigma F_Y &= F_{BC} \sin 60^\circ \\ \Sigma F_Y &= (2 \times 10^{-7})\left(\frac{1}{2}\sqrt{3}\right) = \sqrt{3} \times 10^{-7} \text{ N} \\ \tan \theta &= \frac{\Sigma F_Y}{\Sigma F_X} = \frac{\sqrt{3} \times 10^{-7}}{1 \times 10^{-7}} = \sqrt{3} \\ \theta &= 60^\circ\end{aligned}$$

The result is 60° .

(ii)

$$\vec{F} = I(\vec{l} \times \vec{B})$$

Or The magnitude of the force is $F = I l B \sin \theta$

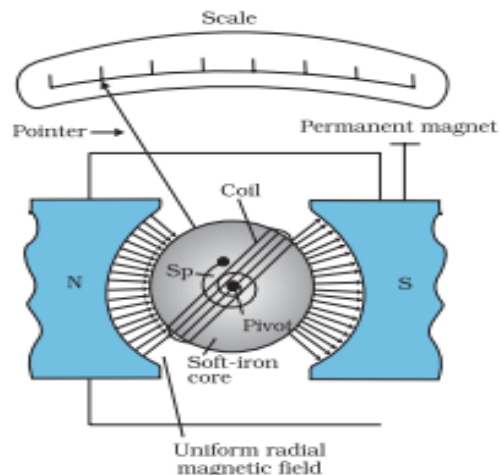
(i) If $\theta = 0^\circ$ or 180° , then $F = I l B(0) = 0$

Thus a current carrying conductor placed parallel to the direction of the magnetic field does not experience any force.

(ii) If $\theta = 90^\circ$, then $F = I l B \sin 90^\circ$ or $F_{\max} = I l B$

Or

PRINCIPLE: It works on the principle that when a current loop is placed in an external magnetic field, it experiences torque, and the value of torque can be changed by changing the current in the loop

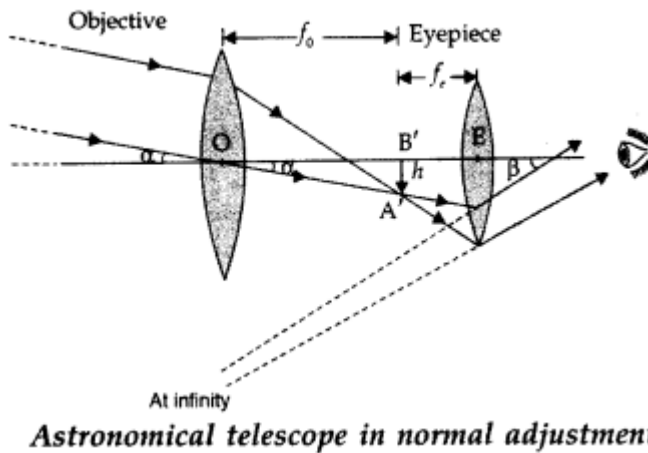


	Theory and working: As the field is radial, the plane of the coil always remains parallel to the field $\vec{B}$. When a current flows through the coil, a torque acts on it. It is $\tau = \text{Force} \times \text{perpendicular distance} = NibB \times a \sin 90 = NIB(ab) = NIBA$ Here $\theta=90^\circ$, because the normal to the plane of coil remains perpendicular to the field $\vec{B}$ in all positions. The torque τ deflects the coil through an angle α. A restoring torque is set up in the coil due to the elasticity of the springs such that $\tau_{\text{restoring}} = k\alpha$ Where k is the torsion constant of the springs i.e., torque required to produce unit angular twist. In equilibrium position, Restoring torque = Deflecting torque $k\alpha = NIBA$ $\alpha = \frac{NBAI}{k}$ Thus the deflection produced in the galvanometer coil is proportional to the current flowing through it. Functions: (i) A uniform magnetic field provides a linear current scale. (ii) A soft iron core makes the field radial. It also increases the strength of the magnetic field and hence increases the sensitivity of the galvanometer.	1/2 1 1 1
32	(a) Correct Ray diagram Correct expression of refractive index of glass in terms of angle of prism and angle of minimum deviation (b)	1 1 1

$$\begin{aligned}
 f_{\text{air}} &= 20 \text{ cm} \\
 {}^a\mu_g &= 1.5 \\
 {}^a\mu_w &= 1.33 \\
 f_w &= ? \\
 \frac{1}{f_{\text{air}}} &= ({}^a\mu_g - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \\
 \frac{1}{20} &= (1.5 - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \\
 \frac{1}{20} &= 0.5 \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \\
 \frac{1}{R_1} - \frac{1}{R_2} &= \frac{1}{20 \times 0.5} = \frac{1}{10} \\
 \text{Now } \frac{1}{f_w} &= ({}^w\mu_g - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \\
 &= \left(\frac{{}^a\mu_g}{{}^a\mu_w} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \\
 &= \left(\frac{1.5}{1.33} - 1 \right) \frac{1}{10} \\
 \frac{1}{f_w} &= \left(\frac{1.5 - 1.33}{1.33} \right) = \frac{1}{10} = \frac{0.17}{1.33 \times 10} \\
 f_w &= \frac{1.33 \times 10}{0.17} = 78.12 \text{ cm}
 \end{aligned}$$

OR

(a) diagram



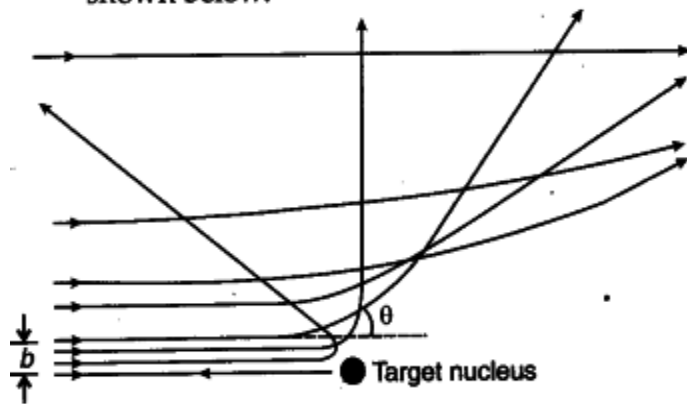
Definition : It is the ratio of the angle subtended at the eye, by the final image, to the angle which the object subtend at the lens, or the eye.

Or Magnifying power $m = -f_o/f_e$

(b) Objective = 0.5 D and eye lens = 10 D

Justification

	(ii) High resolving power/Brighter image/ lower limit of resolution.	
33	(i) According to Bohr's postulates, angular momentum of electron for any permitted orbit is, $mvr = \frac{nh}{2\pi} \quad \text{or} \quad v = \frac{nh}{2\pi mr} \quad \dots(i)$ Also, according to Bohr's postulates, the centripetal force is equal to electrostatic force between the electron and nucleus. $\frac{mv^2}{r} = K \frac{Ze \cdot e}{r^2} \quad \text{or} \quad \frac{m}{r} \cdot \frac{n^2 h^2}{4\pi^2 m^2 r^2} = \frac{KZe^2}{r^2}$ or $r \cdot 4\pi^2 m KZe^2 = n^2 h^2 \therefore r = \frac{n^2 h^2}{4\pi^2 m KZe^2} \quad \dots(ii)$ $r = \frac{h^2}{4\pi m k e^2}$ It is the expression for Bohr's radius, which is about 0.5 Å For first orbit of hydrogen atom, $n = 1$, $Z = 1$ then we have $r = 0.529 \text{ Å}$ (ii) we know that velocity of electron moving around a proton in hydrogen atom in an orbit of radius $5.3 \times 10^{-11} \text{ m}$ is $2.2 \times 10^6 \text{ m/s}$. Thus, the frequency of the electron moving around the proton is $v = \frac{v}{2\pi r} = \frac{2.2 \times 10^6 \text{ ms}^{-1}}{2\pi(5.3 \times 10^{-11} \text{ m})}$ $\approx 6.6 \times 10^{15} \text{ Hz.}$ According to the classical electromagnetic theory we know that the frequency of the electromagnetic waves emitted by the revolving electrons is equal to the frequency of its revolution around the nucleus. Thus the initial frequency of the light emitted is $6.6 \times 10^{15} \text{ Hz}$. OR The trajectory, traced by the α-particles in the Coulomb field of target nucleus,	1 1 1 1 OR 1



The size of the nucleus was estimated by observing the distance (d) of closest approach, of the α -particles. This distance is given by :

$$\frac{1}{4\pi\epsilon_0} \cdot \frac{(Ze)(2e)}{d} = K$$

where K = kinetic energy of the α -particles when they are far away from the target nuclei.

Let d be the distance of closest approach then by the conservation of energy.

Initial kinetic energy of incoming α -particle K

= Final electric potential energy U of the system

As $K = \frac{1}{4\pi\epsilon_0} \times (2e)(Ze)/d$

$$\therefore d = \frac{1}{4\pi\epsilon_0} \cdot \frac{2Ze^2}{K} \quad \text{..(i)}$$

Here,

$$4\pi\epsilon_0 = 9 \times 10^9 \text{ Nm}^2 \text{ C}^{-2}$$

$$Z = 79, e = 1.6 \times 10^{-19} \text{ C.}$$

$$K = 7.7 \text{ MeV}$$

$$= 7.7 \times 10^6 \times 1.6 \times 10^{-19} \text{ J}$$

$$= 1.2 \times 10^{-12} \text{ J}$$

Substituting these values in (i) and solving, we get

$$d = 3 \times 10^{-14} \text{ m}$$

$$m = 30 \text{ fm} (\because 1 \text{ fm} = 10^{-15} \text{ m})$$

END